A Bare Tuple Construction of Near-Ramanujan (q, q^3)-Biregular Graphs

Abstract

We give a completely explicit bipartite graph defined only using tuples over a finite field. It is (q, q^3)-biregular and its nontrivial normalized singular values are at most

$$\sqrt{\frac{2q^2 - q + 1}{q^3}} = \sqrt{\frac{2}{q} + O(q^{-2})}.$$

Equivalently, if $\Delta = q^3$ is the larger degree, then its unnormalized second singular value is at most $\sqrt{2\Delta + O(q^2)}$. The proof is a direct Fourier calculation plus a finite-field point count.

1 The graph

Let $\mathbb{F} = \mathbb{F}_q$. Let

$$\mathcal{V} = \mathbb{F}^8.$$

We write an element $v \in \mathcal{V}$ as

$$v = (v_{11}, v_{12}, v_{13}, v_{21}, v_{22}, v_{23}, v_{31}, v_{32}).$$

The subscripts are just names for the coordinates. They are meant to suggest the traceless matrix

$$\begin{pmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & -v_{11} - v_{22} \end{pmatrix},$$

but no matrix language is needed for the construction.

For each triple $(x, y, z) \in \mathbb{F}^3$, define a direction vector

$$\rho(x, y, z) = \big(z, -xz, xyz - z^2, y, -xy, xy^2 - yz, 1, -x\big) \in \mathcal{V}.$$

The important feature is that its seventh coordinate is always 1.

Define a bipartite graph $Y_q = (L, R, E)$ as follows.

The right side is

$$R = \mathcal{V}.$$

The left side consists of triples (x, y, z) together with an affine line in $\mathcal{V}$ parallel to $\rho(x, y, z)$. To make this literally a set of tuples, we use the unique representative of this line whose seventh coordinate is 0. Thus

$$L = \big\{(x, y, z, w) : x, y, z \in \mathbb{F},\ w \in \mathcal{V},\ w_{31} = 0\big\}.$$

A right vertex $v \in R$ is adjacent to a left vertex $(x, y, z, w) \in L$ if and only if

$$v = w + a\rho(x, y, z) \quad \text{for some } a \in \mathbb{F}.$$

Equivalently, the left vertex (x, y, z, w) represents the q-point affine line

$$w + \mathbb{F}\rho(x, y, z) \subseteq \mathcal{V},$$

and it is adjacent exactly to the points of that line.

Lemma 1 (Degrees)**.** *The graph Y_q is (q, q^3)-biregular: every left vertex has degree q, and every right vertex has degree q^3.*

Proof. A left vertex is an affine line with q points, so its degree is q.

Now fix a right vertex $v \in \mathcal{V}$. For each $(x, y, z) \in \mathbb{F}^3$, there is exactly one line of direction $\rho(x, y, z)$ through v. Because the seventh coordinate of $\rho(x, y, z)$ is 1, the unique representative w of this line with $w_{31} = 0$ is

$$w = v - v_{31}\rho(x, y, z).$$

Thus v has exactly one neighbor for each $(x, y, z) \in \mathbb{F}^3$, hence degree q^3. □

The rest of the note proves the spectral bound.

2 Squaring the graph

Let A be the normalized bipartite adjacency operator from functions on R to functions on L:

$$(Af)(x, y, z, w) = \frac{1}{q} \sum_{a \in \mathbb{F}} f\big(w + a\rho(x, y, z)\big).$$

The nonzero singular values of A are the square-roots of the eigenvalues of the two-step operator

$$T = A^* A$$

on functions on $R = \mathcal{V}$.

Starting from $v \in \mathcal{V}$, the two-step walk first chooses a direction $(x, y, z) \in \mathbb{F}^3$, then chooses a random point on the corresponding line through v. Therefore

$$(Tf)(v) = \frac{1}{q^4} \sum_{x,y,z \in \mathbb{F}} \sum_{a \in \mathbb{F}} f\big(v + a\rho(x, y, z)\big).$$

So T is a translation-invariant averaging operator on the additive group $\mathcal{V}$.

Fix a nontrivial additive character $\psi : \mathbb{F} \to \mathbb{C}^\times$. For every linear functional $\lambda : \mathcal{V} \to \mathbb{F}$, define

$$\chi_\lambda(v) = \psi(\lambda(v)).$$

These characters form an orthonormal basis for $L^2(\mathcal{V})$. The character χ_0 is the constant function.

For a character χ_λ,

$$T\chi_\lambda = \eta(\lambda)\chi_\lambda,$$

where

$$\eta(\lambda) = \frac{1}{q^4} \sum_{x,y,z \in \mathbb{F}} \sum_{a \in \mathbb{F}} \psi\big(a\lambda(\rho(x, y, z))\big).$$

Averaging over a gives

$$\eta(\lambda) = \frac{1}{q^3} \#\big\{(x, y, z) \in \mathbb{F}^3 : \lambda(\rho(x, y, z)) = 0\big\}.$$

Thus the spectral problem is exactly a point-counting problem.

3 The point count

As λ ranges over all linear functionals on $\mathcal{V}$, the polynomial

$$F_\lambda(x,y,z) = \lambda(\rho(x,y,z))$$

ranges over the following 8-dimensional space of polynomials:

$$\mathcal{P} = \left\{ \begin{aligned} F(x,y,z) =& a + bx + cy + dz + exy + fxz \\ &+ g(xy^2 - yz) + h(xyz - z^2) : a,b,c,d,e,f,g,h \in \mathbb{F} \end{aligned} \right\}.$$

The zero polynomial corresponds exactly to the trivial character.

Lemma 2 (Point count)**.** *If $F \in \mathcal{P}$ is nonzero, then*

$$\#\{(x,y,z) \in \mathbb{F}^3 : F(x,y,z) = 0\} \leq 2q^2 - q + 1.$$

The bound is sharp.

Proof. Write

$$F(x,y,z) = A(y,z) + xB(y,z),$$

where

$$A(y,z) = a + cy + dz - gyz - hz^2$$

and

$$B(y,z) = b + ey + fz + gy^2 + hyz.$$

For fixed (y,z), the equation

$$A(y,z) + xB(y,z) = 0$$

has exactly one solution in x if $B(y,z) \neq 0$, has q solutions if $A(y,z) = B(y,z) = 0$, and has no solutions if $B(y,z) = 0$ but $A(y,z) \neq 0$. Hence

$$|Z(F)| = q^2 - |Z(B)| + q|Z(A,B)|,$$

where

$$Z(B) = \{(y,z) : B(y,z) = 0\}, \quad Z(A,B) = \{(y,z) : A(y,z) = B(y,z) = 0\}.$$

If $B = 0$, then $F = A$ is a nonzero affine-linear function of (y,z), so

$$|Z(F)| \leq q^2.$$

This is enough.

Assume $B \neq 0$. If $|Z(B)| \leq q$, then

$$|Z(F)| \leq q^2 + (q-1)|Z(B)| \leq q^2 + (q-1)q = 2q^2 - q.$$

It remains to consider $|Z(B)| > q$. Since

$$B(y,z) = b + ey + fz + gy^2 + hyz$$

is linear in z, this can only happen in the following elementary cases: its zero set is the union of two affine lines in the (y,z)-plane, and

$$|Z(B)| \geq 2q - 1.$$

For $q \geq 3$, a degree-2 polynomial on an affine line that vanishes at all q points of that line vanishes identically on the line. Thus, if A vanished on both line components of $Z(B)$, then the quadratic polynomial B would divide A. This is impossible by comparing the y^2, yz, and z^2 coefficients in the displayed forms of A and B. Hence A vanishes identically on at most one component of $Z(B)$. If it vanishes on one component, then after dividing by that line the remaining factor is linear, and hence it has at most one zero on the other component. If it vanishes on no component, Bezout gives at most four intersection points, which is at most $q+1$. In all cases

$$|Z(A,B)| \leq q+1.$$

Consequently

$$|Z(F)| \leq q^2 - (2q-1) + q(q+1) = 2q^2 - q + 1.$$

For $q = 2$, the same bound says that a nonzero $F \in \mathcal{P}$ cannot vanish on all eight points of $\mathbb{F}_2^3$. Here this is checked directly. If F vanishes everywhere, then evaluating at

$$(0,0,0), (1,0,0), (0,1,0), (0,0,1), (1,1,0), (1,0,1), (0,1,1), (1,1,1)$$

forces successively

$$a = b = c = e = f = g = d = h = 0.$$

Thus a nonzero F has at most seven zeros, which is $2q^2 - q + 1$ when $q = 2$.

Finally, the bound is sharp. Choose distinct $s, t \in \mathbb{F}$ and take

$$F_{s,t}(x,y,z) = (z-s)(xy - z + t).$$

This belongs to $\mathcal{P}$, since

$$F_{s,t} = xyz - z^2 - sxy + (s+t)z - st.$$

Its zero set is the union of the plane $z = s$ and the surface $xy = z - t$. Counting without double-counting the intersection gives

$$|Z(F_{s,t})| = 2q^2 - q + 1.$$

□

4 Spectrum

Theorem 1. *All nontrivial normalized singular values of Y_q are at most*

$$\sqrt{\frac{2q^2 - q + 1}{q^3}}.$$

Moreover this bound is attained. Thus

$$\sigma_2(Y_q) = \sqrt{\frac{2q^2 - q + 1}{q^3}} = \sqrt{\frac{2}{q} - \frac{1}{q^2} + \frac{1}{q^3}}.$$

Proof. The nonzero singular values of the normalized bipartite adjacency operator are the square-roots of the eigenvalues of $T = A^*A$. The characters χ_λ diagonalize T, and for nontrivial λ,

$$\eta(\lambda) = \frac{|Z(F_\lambda)|}{q^3}.$$

By the point-counting lemma,

$$\eta(\lambda) \le \frac{2q^2 - q + 1}{q^3}.$$

Taking square-roots gives the asserted upper bound.

The sharpness example in the point-counting lemma comes from a nonzero linear functional, so the corresponding character gives an eigenvalue exactly

$$\frac{2q^2 - q + 1}{q^3}.$$

Thus the bound is attained. □

Since Y_q is (q, q^3)-biregular, the unnormalized second singular value is

$$\lambda_2(Y_q) = \sqrt{q \cdot q^3}\, \sigma_2(Y_q) = q^2 \sqrt{\frac{2q^2 - q + 1}{q^3}} = \sqrt{2q^3 - q^2 + q}.$$

If $\Delta = q^3$ is the larger degree, this is

$$\lambda_2(Y_q) = \sqrt{2\Delta + O(q^2)} = O(\sqrt{\Delta}).$$

Remark 1. *The construction above is the bare tuple-level version of the simplest KMS/KO shadow graph. It intentionally avoids cosets, buildings, and complexes: the vertices are just vectors and affine lines in the explicitly parametrized directions $\rho(x, y, z)$.*